

Lecture 10 : Class Exercise Sheet

	2025	2024	
Current Assets			
Cash	1000	500	
Accounts Receivables	400	500	
Inventory	600	900	
Total	2000	1900	
Fixed Assets			
Land and Buildings	1200	1050	
Machinery and Equipment	850	800	
Furniture and Fixtures	300	220	
Vehicles	100	80	
Leases	50	50	
Total fixed assets	2500	2200	
Less: Accumulated Depreciation	(1300)	(1200)	
Net Fixed Assets	1200	1000	
Total Assets	3200	2900	
Current Liabilities			
Accounts Payable	700	500	
Notes Payable	600	700	
Accruals	100	200	
Total Current liabilities	1400	1400	
Long term debt	600	400	
Total Liabilities	2000	1800	
Shareholder's Equity			
Capital Stock	220	220	
Paid in Capital	380	380	
Retained Earnings	600	500	
Total Stockholder's Equity	1200	1100	
Liabilities and Stockholder's Equity	3200	2900	

Task One: The above is a consecutive balance sheet for two years for Thomson Inc.

Additional Information for the year 2025::

Depreciation Expense for 2025 is \$100

Earnings before interest and taxed \$370

Interest Expense \$70

Tax rate =40%

Net income = \$180

Calculating Free Cash Flows:

Free Cash Flow (FCF) is the amount of cash flow available to debt and equity holders after meeting all operating (OCF) needs and paying for its **net fixed asset investments (NFAI)** and **net current asset investments (NCAI)**.

$$\text{Free Cash Flows (FCF)} = \text{OCF} - \text{NFAI} - \text{NCAI}$$

$$1\text{-NOPAT } 2024 = [\text{EBIT} (1 - T)]$$

2- Total operating Capital=

$$\begin{aligned} & \text{Operating working Capital} + \text{Net Fixed Assets Investment} \\ = & [\text{Operating Current Assets} - \text{Operating current liabilities}] \end{aligned}$$

2024=

2025=

3- Net investment in operating capital (difference between 2024 and 2025)

$$4\text{-Free Cash Flows (FCF)} = \text{NOPAT} - \text{Net investment in operating capital}$$

Task Two:

Magna Fax, Inc. Income Statement For the Year Ended December 31, 2024		
Sales revenue		\$150,000
Cost of goods sold		<u>117,500</u>
Gross Profits		\$32,500
Selling expense	4,500	
General and administrative expense	4,000	
Depreciation expense	<u>4,000</u>	
Operating profits EBIT		\$ 20,000
Interest expense		<u>2,500</u>
Net profit before taxes EBT		\$ 17,500
Taxes (40%)		<u>7,000</u>
Net profit after taxes		\$ 10,500

Magna Fax, Inc. Balance Sheet For the Years Ended December 31, 2023 and 2024				
		2024		2023
Assets				
Cash		\$24,000		\$21,000
Accounts receivable		45,000		39,000
Inventory		30,000		27,000
Gross fixed assets	\$42,000		\$40,000	
Acc. Depreciation	<u>(22,000)</u>		<u>(18,000)</u>	
Net fixed assets		<u>20,000</u>		<u>22,000</u>
Total assets		\$119,000		\$109,000
Liabilities and Equity				
Accounts payable		\$25,000		\$30,000
Notes payable		50,000		40,000
Accruals		1,000		2,000
Long-term debts		10,000		8,000
Common stock at par		1,000		1,000
Paid-in capital in excess of par		4,000		4,000
Retained earnings		<u>28,000</u>		<u>24,000</u>
Total liabilities and equity		\$119,000		\$109,000

The credit manager at First National Bank has just received the income statement and balance sheet for Magna Fax, Inc. for the year ended December 31, 2024. The bank requires the firm to report on its earnings performance and financial position quarterly as a condition of a loan agreement. The bank's credit manager must prepare two key financial statements based on the information sent by Magna Fax, Inc. This will be passed on to the commercial loan officer assigned to this account, so that he may review the financial condition of the firm.

Required: Calculate the company's free cash flows for 2024

Multiple Choice Questions

1-EP Enterprises has the following income statement. How much net operating profit after taxes (NOPAT) does the firm have?

Sales	\$1,800.00
Costs	1,400.00
Depreciation	<u>250.00</u>
EBIT	\$ 150.00
Interest expense	<u>70.00</u>
EBT	\$ 80.00
Taxes (40%)	<u>32.00</u>
Net income	<u><u>\$ 48.00</u></u>

- a. \$81.23
- b. \$85.50
- c. \$90.00
- d. \$94.50
- e. \$99.23

2- Tibbs Inc. had the following data for the year ending 12/31/22: Net income = \$300; Net operating profit after taxes (NOPAT) = \$400; Total assets = \$2,500; Short-term investments = \$200; Stockholders' equity = \$1,800; Total debt = \$700; and Total operating capital = \$2,300. What was its return on invested capital (ROIC)?

- a. 14.91%
- b. 15.70%
- c. 16.52%
- d. 17.39%
- e. 18.26%

3. Zumbahlen Inc. has the following balance sheet. How much total operating capital does the firm have?

Cash	\$ 20.00	Accounts payable	\$ 30.00
Short-term investments	50.00	Accruals	50.00
Accounts receivable	20.00	Notes payable	<u>30.00</u>
Inventory	<u>60.00</u>	Current liabilities	\$110.00
Current assets	\$150.00	Long-term debt	70.00
Gross fixed assets	\$140.00	Common stock	30.00
Accumulated deprec.	<u>40.00</u>	Retained earnings	<u>40.00</u>
Net fixed assets	<u>\$100.00</u>	Total common equity	<u>\$ 70.00</u>
Total assets	<u>\$250.00</u>	Total liab. & equity	<u>\$250.00</u>

- a. \$114.00
- b. \$120.00
- c. \$126.00
- d. \$132.30
- e. \$138.92

4- HHH Inc. reported \$12,500 of sales and \$7,025 of operating costs (including depreciation). The company had \$18,750 of investor-supplied operating assets (or capital), the weighted average cost of that capital (the WACC) was 9.5%, and the federal-plus-state income tax rate was 40%. What was HHH's Economic Value Added (EVA), i.e., how much value did management add to stockholders' wealth during the year?

- a. \$1,357.13
- b. \$1,428.56
- c. \$1,503.75
- d. \$1,578.94
- e. \$1,657.88